Markov-switching GARCH and the persistence puzzle

Summaries of the core papers for the master's thesis

Reading guide · 23 September 2026

Contents

1 How to use this guide **3**

2 Foundations: from ARCH to the persistence puzzle (Chapter 2) **5**

2.1 Engle (1982): the ARCH model . . . 5
2.2 Bollerslev (1986): the GARCH model . . . 5
2.3 Bollerslev (1987): Student-t GARCH . . . 6
2.4 Bollerslev & Engle (1993): persistence and co-persistence . . . 7
2.5 Diebold (1986): comment on integrated variance . . . 7
2.6 Lamoureux & Lastrapes (1990a): volume versus GARCH . . . 8
2.7 Hamilton (1989): Markov switching and the filter . . . 8

3 The models and the path dependence problem (Chapter 3) **10**

3.1 The problem in one box . . . 10
3.2 Cai (1994): Markov-switching ARCH . . . 10
3.3 Hamilton & Susmel (1994): SWARCH . . . 11
3.4 Gray (1996): generalized regime switching . . . 12
3.5 Klaassen (2002): regime-switching GARCH forecasts . . . 12
3.6 Kim (1994): switching state-space models and collapsing . . . 13
3.7 Francq, Roussignol & Zakoïan (2001): GARCH driven by a hidden Markov chain . . . 14
3.8 Francq & Zakoïan (2005): moments and ARMA structure . . . 14
3.9 Bauwens, Preminger & Rombouts (2010): Bayesian MS-GARCH . . . 15
3.10 Augustyniak (2014): maximum likelihood for the path-dependent model . . . 15
3.11 Haas, Mittnik & Paolella (2004): the focal model . . . 16
3.12 Haas & Liu (2015): theory and a multivariate extension . . . 17

4 Theory for IS-GARCH (Chapter 4) **19**

4.1 Nelson (1990): stationarity and persistence of GARCH(1,1) . . . 19
4.2 Bougerol & Picard (1992): the Lyapunov-exponent condition . . . 19
4.3 Liu (2006): stationarity of IS-GARCH . . . 20
4.4 Francq & Zakoïan (2004): QMLE for GARCH . . . 20
4.5 Newey & Steigerwald (1997): consistency of non-Gaussian QMLE . . . 21
4.6 Davies (1977): nuisance parameters under the alternative . . . 21
4.7 Cho & White (2007): testing for regime switching . . . 21

5 Software (Chapter 5) **22**

5.1 Ardia et al. (2019): the MSGARCH package . . . 22

6 Synthesis: where persistence went **23**

7 Open issues raised by the reading **23**

References **24**

1 How to use this guide

This guide summarises the 26 papers in the project folder that the thesis will rely on. It is a reading aid, not thesis text: it uses lists freely, and its sentences are paraphrases written for your understanding rather than for citation. When a point makes its way into a chapter, it will be rewritten and cited from the paper itself.

Notation. Every equation is written in the thesis notation (`notation.md`), not the paper's own symbols. So S_t is always the regime, $\sigma^2_{k,t}$ the regime-k variance, $p_{ij} = P(S_t = j \mid S_{t-1} = i)$, and $\xi_{k,t|t}$ the filtered probability, whatever the original paper wrote. Where a paper's convention differs in a way that could confuse you when you open the PDF, the summary says so.

Numbers. Every number reported here was checked against the paper; numbers read from scanned tables were checked against the page image. Figures that I computed myself (for example expected durations $1/(1-p_{kk})$ or sums $\alpha+\beta$) are marked *(computed)*. Anything not fully verified is marked [VERIFY].

Structure of each summary. Bibliographic line; where it is used in the thesis; the question the paper asks; the model or method (in our notation); the main results; and a box explaining why it matters for the thesis, including any caution it raises.

The project in one paragraph

Single-regime GARCH models of daily returns typically estimate a persistence $\alpha+\beta$ close to one, meaning volatility shocks fade very slowly. One explanation is that this persistence is partly an artifact: if volatility occasionally shifts between regimes with different levels, a model that ignores the shifts attributes their persistence to individual shocks. Markov-switching GARCH models let persistence live in two places, the regime process ($p_{11}+p_{22}-1$) and the within-regime GARCH dynamics ($\alpha_k+\beta_k$). Combining them, however, creates *path dependence*: the variance depends on the whole unobserved regime history, so the likelihood is intractable. The literature can be read as four responses: *avoid* it (use ARCH instead of GARCH), *collapse* it (replace the path-dependent variance with an average), *confront* it (keep the exact model and use simulation methods), or *remove* it (redefine the model so that each regime has its own GARCH recursion). The thesis compares these responses on paper and on daily S&P 500 returns, with Haas, Mittnik and Paolella's (2004) model as the focal specification.

Map of papers to chapters

Paper	Chapter	Role
Engle (1982)	2.1	ARCH; origin of conditional variance modelling
Bollerslev (1986)	2.2	GARCH; defines persistence $\alpha+\beta$
Bollerslev (1987)	2.2, 2.3	Student-t innovations; near-unit persistence in daily data
Bollerslev & Engle (1993)	2.3	IGARCH and persistence in variance (file mislabelled)
Diebold (1986)	2.3	integrated variance may reflect omitted regime shifts
Lamoureux & Lastrapes (1990a)	2.3	persistence vanishes once volume is included
Hamilton (1989)	2.4, Ch. 4	Markov switching and the filter
Cai (1994)	3.2	Branch 1: MS-ARCH with switching intercept
Hamilton & Susmel (1994)	2.3, 3.2	Branch 1: SWARCH
Gray (1996)	3.3	Branch 2: collapse by averaging over regimes
Klaassen (2002)	3.3	Branch 2: improved collapse; forecasting
Kim (1994)	3.3, Ch. 4	collapsing idea; smoother
Francq, Roussignol & Zakoïan (2001)	3.2, 3.4	Branch 3 theory; MS-ARCH consistency
Francq & Zakoïan (2005)	3.4	Branch 3: moments and ARMA structure

Paper	Chapter	Role
Bauwens, Preminger & Rombouts (2010)	3.4	Branch 3: Bayesian estimation; S&P 500
Augustyniak (2014)	3.3, 3.4	Branch 3: exact ML by simulation; S&P 500
Haas, Mittnik & Paolella (2004)	2.4, 3.5, Ch. 4	Branch 4: the focal IS-GARCH model
Haas & Liu (2015)	3.5, Ch. 4, (6)	Branch 4: theory, MLE consistency; multivariate
Nelson (1990)	Ch. 4	strict stationarity of GARCH(1,1); IGARCH
Bougerol & Picard (1992)	Ch. 4	Lyapunov-exponent condition
Liu (2006)	Ch. 4	stationarity theory for IS-GARCH
Francq & Zakoïan (2004)	Ch. 4	QMLE asymptotics for GARCH
Newey & Steigerwald (1997)	Ch. 4	consistency of non-Gaussian QMLE
Davies (1977)	Ch. 4	nuisance parameters under the alternative
Cho & White (2007)	Ch. 4	testing one versus two regimes
Ardia et al. (2019)	Ch. 5	the MSGARCH R package

Suggested reading order

1. **The basics:** Engle (1982), Bollerslev (1986), Hamilton (1989).
2. **The puzzle:** Diebold (1986), Bollerslev (1987), Bollerslev & Engle (1993), Lamoureux & Lastrapes (1990a).
3. **The problem:** the box on path dependence (Section 3.1), then Cai (1994) and Hamilton & Susmel (1994).
4. **The focal model:** Haas et al. (2004), then Gray (1996) and Klaassen (2002) to see what it reacts against.
5. **The S&P 500 evidence:** Bauwens et al. (2010) and Augustyniak (2014).
6. **Theory for Chapter 4:** Nelson (1990), Bougerol & Picard (1992), Liu (2006), Francq & Zakoïan (2004), Haas & Liu (2015), Newey & Steigerwald (1997).
7. **Testing:** Davies (1977), Cho & White (2007).
8. **Reference as needed:** Kim (1994), Francq et al. (2001), Francq & Zakoïan (2005), Ardia et al. (2019).

If you only have time for five papers, read Hamilton (1989), Hamilton & Susmel (1994), Haas et al. (2004), Augustyniak (2014) and Haas & Liu (2015). The synthesis table in Section 6 puts the persistence evidence from all papers side by side.

2 Foundations: from ARCH to the persistence puzzle (Chapter 2)

2.1 Engle (1982): the ARCH model

Engle, R. F. (1982). Autoregressive conditional heteroscedasticity with estimates of the variance of United Kingdom inflation. *Econometrica*, 50(4), 987–1008.

Used in: §2.1. Note that the PDF is an image-only scan.

Question. Econometric models assumed that the one-period forecast variance is constant. Engle asks whether the variance of a forecast can itself be predicted from the recent past, and how to estimate and test such a model.

Model. For the innovation of a mean or regression equation,

$$\varepsilon_t = \sigma_t \eta_t, \qquad \sigma_t^2 = \omega + \sum_{m=1}^{q} a_m \varepsilon_{t-m}^2, \qquad \eta_t \sim \text{i.i.d. } N(0,1).$$

The process is serially uncorrelated with a constant unconditional variance, but its conditional variance moves with recent squared shocks.

Main results.

- Covariance stationarity holds if and only if the roots of the characteristic equation associated with the a_m lie outside the unit circle; the unconditional variance is then $\omega/(1-\sum a_m)$ (Theorem 2).
- Even the first-order model has a fourth moment above the normal value whenever $a_1 \neq 0$: conditional normality plus time-varying variance gives fat unconditional tails.
- Maximum likelihood is more efficient than least squares, which nonetheless remains consistent.
- **LM test for ARCH:** regress squared residuals on a constant and q of their lags; TR^2 is asymptotically χ_q^2 under the null.

Evidence. Quarterly UK consumer price inflation, 1958:II–1977:II. The LM test for first-order ARCH was not significant, but for fourth-order ARCH it was ($\chi_4^2 = 15.2$). Engle imposed fixed, linearly declining weights (0.4, 0.3, 0.2, 0.1) on four lagged squared residuals, because an unrestricted version would probably violate non-negativity and stationarity. The implied standard deviation of inflation rose from about 0.6% in the late 1960s to about 1.5% after 1974.

Why it matters for the thesis

The starting point of the whole literature. The detail worth remembering is the *imposed lag shape*: volatility seemed to depend on a long history of shocks, and ARCH could only capture that by fixing the weights in advance. That is exactly the problem GARCH solves, and it is the first hint that volatility memory is long. The LM test is also a standard residual diagnostic you can use in Chapter 5.

2.2 Bollerslev (1986): the GARCH model

Bollerslev, T. (1986). Generalized autoregressive conditional heteroskedasticity. *Journal of Econometrics*, 31, 307–327.

Used in: §2.2; the single-regime benchmark in Chapter 5.

Question. Early ARCH applications needed long, arbitrarily restricted lag structures. Can a more flexible, parsimonious specification do the same job?

Model. GARCH(p, q) adds lagged conditional variances. The workhorse is

$$\sigma_t^2 = \omega + \alpha \varepsilon_{t-1}^2 + \beta \sigma_{t-1}^2, \qquad \omega > 0,\ \alpha, \beta \geq 0.$$

The analogy is AR → ARMA: a short recursion replaces a long lag distribution. If $\beta < 1$ it can be written as ARCH(∞):

$$\sigma_t^2 = \frac{\omega}{1-\beta} + \alpha \sum_{m=1}^{\infty} \beta^{m-1} \varepsilon_{t-m}^2.$$

Main results.

- **Theorem 1:** wide-sense stationarity if and only if $\alpha + \beta < 1$ (in general, the sum of all ARCH and GARCH coefficients < 1); the unconditional variance is $\bar{\sigma}^2 = \omega/(1-\alpha-\beta)$.
- ε_t^2 has an ARMA representation, and beyond the first lag its autocorrelations decay geometrically at rate $\alpha + \beta$ (a Yule–Walker analogue useful for identification).
- Conditions for higher moments; GARCH(1,1) is leptokurtic.
- The median lag of the variance equation is $-\ln 2 / \ln \beta$ and the mean lag is $1/(1-\beta)$.

Evidence. Quarterly US GNP-deflator inflation, 1948:2–1983:4 (143 observations). GARCH(1,1) gives $\hat{\alpha} = 0.135$, $\hat{\beta} = 0.829$ (so $\alpha + \beta \approx 0.96$, *computed*). It fits slightly better than an eighth-order ARCH with fixed declining weights, with an estimated mean lag of 5.85 and median lag of 3.70 quarters.

Why it matters for the thesis

Defines the benchmark model and the quantity at the centre of the thesis. Keep in mind the three roles of $\alpha + \beta$: stationarity boundary, decay rate of the autocorrelations of ε_t^2, and decay rate of variance forecasts (the last via Hamilton & Susmel's forecast equation). A single fixed number controls all three. Regime-switching models relax precisely this.

2.3 Bollerslev (1987): Student-t GARCH

Bollerslev, T. (1987). A conditionally heteroskedastic time series model for speculative prices and rates of return. *The Review of Economics and Statistics*, 69(3), 542–547.

Used in: §2.2 (Student-t), §2.3 (daily vs monthly persistence).

Question. Conditionally normal GARCH produces fat tails, but are they fat enough for asset returns?

Model. GARCH(1,1) with η_t following a Student-t with ν degrees of freedom, scaled so that the conditional variance is σ_t^2 (this requires $\nu > 2$). As $\nu \to \infty$ it becomes the normal model; Bollerslev estimates $1/\nu$. He also shows the model can be derived as a subordinated process with an unobserved error in the variance equation.

Evidence (Table 2). Two samples:

- Daily US-dollar exchange rates, March 1980–January 1985 (1,245 observations):
  - Sterling: $\alpha = 0.057$, $\beta = 0.921$ ($\alpha + \beta = 0.978$, *computed*), $\nu \approx 8$.
  - Deutschmark: $\alpha = 0.095$, $\beta = 0.881$ (0.976), $\nu \approx 14$.
- Five *monthly* US stock price indices, including the S&P 500 Composite: $\alpha = 0.074$, $\beta = 0.768$ (0.842), $1/\nu = 0.139$ so $\nu \approx 7.2$ *(computed)*.

The likelihood-ratio statistics against normality range from 10.46 to 41.26, well above χ_1^2 critical values. Bollerslev notes that $\alpha + \beta$ is close to one for both currencies and suggests an integrated GARCH-t might suit them.

Why it matters for the thesis

Source for Student-t innovations, the distribution of the focal model. The table also carries a useful nuance for §2.3: near-unit persistence shows up in *daily* data, while the monthly S&P 500 estimate is 0.84. Persistence is partly a property of sampling frequency, not a universal constant.

2.4 Bollerslev & Engle (1993): persistence and co-persistence

Bollerslev, T., & Engle, R. F. (1993). Common persistence in conditional variances. *Econometrica*, 61(1), 167–186.

Used in: §2.3. **Note:** the project file is labelled "Engle_and_Bollerslev_1986" but contains this 1993 paper. The original IGARCH paper (Engle & Bollerslev, 1986) is not in the folder; cite it as a secondary source through this paper.

Question. High-frequency financial data often show an approximate unit root in the conditional variance (the IGARCH case of Engle & Bollerslev, 1986). If several series are persistent in variance, can a combination of them be non-persistent, in analogy with co-integration?

Key ideas.

- **Persistence in variance:** a shock to the conditional variance remains important for forecasts at all horizons. For univariate GARCH(1,1) this is the IGARCH case $\alpha + \beta = 1$.
- **Co-persistence:** a linear combination of variance-persistent series may have no persistence, so the series share a common long-run volatility component. Necessary and sufficient conditions are given for the linear multivariate GARCH model.

Evidence. Daily Deutschmark and sterling against the US dollar, January 1980–February 1985 (1,245 observations). The bivariate system looks persistent in variance, but the Deutschmark/sterling cross rate shows no persistence. This suggests that most of the persistent volatility is dollar-related news.

Why it matters for the thesis

Your citable source for the IGARCH idea and the "approximate unit root" stylised fact. It also offers a rival answer to "where does persistence live": in a *common factor* rather than in regimes. That is worth one sentence in §2.3 to show the regime explanation is not the only one.

2.5 Diebold (1986): comment on integrated variance

Diebold, F. X. (1986). Modeling the persistence of conditional variances: A comment. *Econometric Reviews*, 5(1), 51–56.

Used in: §2.3. A six-page comment on Engle & Bollerslev (1986).

Main points.

- **Statistical motivation:** the Wold decomposition only requires the *unconditional* variance to be constant, so GARCH is compatible with covariance stationarity. He singles out GARCH with Student-t innovations as the most useful contribution.
- **Economic motivation:** if the data are sampled in calendar time while the economy evolves in "information time", GARCH-type clustering can appear. This is the time-deformation argument later tested by Lamoureux & Lastrapes.
- GARCH undermines standard serial-correlation tests.
- **Integrated variance is "dangerous":**
  - unit roots in the variance are hard to distinguish from trends;

  - initial conditions matter;
  - persistent variance movements call for an economic explanation. In particular, apparently integrated variance may reflect a failure to include regime dummies in the intercept ω, giving stationary GARCH within regimes and jumps between them.
- Temporal aggregation of GARCH processes pushes the unconditional distribution towards normality.

Why it matters for the thesis

The seed of the thesis framing. Diebold's regime-dummy remark is the earliest statement in the folder that near-unit persistence might be an artifact of unmodelled regime shifts in ω. Cai (1994) and Haas et al. (2004) both cite it as their starting point.

2.6 Lamoureux & Lastrapes (1990a): volume versus GARCH

Lamoureux, C. G., & Lastrapes, W. D. (1990a). Heteroskedasticity in stock return data: Volume versus GARCH effects. *The Journal of Finance*, 45(1), 221–229.

Used in: §2.3. **Caution:** this is *not* the structural-change paper (Lamoureux & Lastrapes, 1990b, *JBES*) that the MS-GARCH literature usually cites. That one is not in the folder.

Question. Is ARCH in daily stock returns a symptom of serially correlated information arrival (the mixture-of-distributions hypothesis)?

Model. GARCH(1,1) for daily returns with contemporaneous trading volume V_t added to the variance equation as a proxy for information arrival: $\sigma_t^2 = \omega + \alpha\varepsilon_{t-1}^2 + \beta\sigma_{t-1}^2 + cV_t$. (Here c is only a label for this summary, not a thesis symbol.)

Evidence. Daily returns on 20 actively traded US stocks.

- Without volume, the mean $\alpha + \beta$ is 0.728 (median 0.807), with large dispersion across stocks.
- With volume, the volume coefficient is significantly positive for every stock, and the mean $\alpha + \beta$ falls to 0.073 (median 0.057). ARCH effects remain for only four companies.
- The authors acknowledge that volume may be endogenous, and their instruments were weak.

Why it matters for the thesis

Supports the general claim that GARCH persistence can reflect *omitted structure*, here the information flow, rather than genuine shock memory. It does *not* support the regime-shift claim. Two further cautions: the persistence in their no-volume sample is not uniformly near one, and the result concerns individual stocks, not an index. Keep the wording in §2.3 precise.

2.7 Hamilton (1989): Markov switching and the filter

Hamilton, J. D. (1989). A new approach to the economic analysis of nonstationary time series and the business cycle. *Econometrica*, 57(2), 357–384.

Used in: §2.4; the filter underlies every regime model in Chapters 3–5.

Question. Can occasional discrete shifts in the growth rate of output be modelled, estimated and dated when the shifts are not observed?

Model. The mean of an autoregression depends on an unobserved regime $S_t \in \{1, 2\}$ that follows a first-order Markov chain with transition probabilities p_{ij}. The regime is inferred with a nonlinear filter, which is the discrete-state analogue of the Kalman filter. In thesis notation, the filter step is

$$\xi_{k,t|t} = \frac{f(r_t \mid S_t = k, \mathcal{F}_{t-1})\, \xi_{k,t|t-1}}{\sum_j f(r_t \mid S_t = j, \mathcal{F}_{t-1})\, \xi_{j,t|t-1}},$$

the prediction step is

$$\xi_{k,t+1|t} = \sum_i p_{ik}\, \xi_{i,t|t},$$

and the log-likelihood is

$$\ell_T(\theta) = \sum_t \ln \sum_k f(r_t \mid S_t = k, \mathcal{F}_{t-1})\, \xi_{k,t|t-1}.$$

In Hamilton's model the density depends on a few lagged regimes as well, so the filter runs over joint configurations of current and lagged regimes. This works only because the number of relevant lags is *finite*, which is the point that later rules out GARCH. A smoother gives $\xi_{k,t|T}$.

Evidence. Quarterly US real GNP growth, 1951:II–1984:IV.

- The likelihood prefers business-cycle regimes over decade-long trend regimes: about -0.4% growth per quarter in one regime and $+1.2\%$ in the other.
- Staying probabilities are 0.905 (expansion) and 0.755 (recession), so expected durations are about 10.5 and 4.1 quarters *(computed as* $1/(1-p_{kk})$*)*.
- Inferred recession probabilities match NBER turning points closely.
- A typical recession implies a permanent drop of about 3% in the level of GNP.

Why it matters for the thesis

Provides the filter, the smoother, the duration formula $1/(1-p_{kk})$ and the idea of regime *dating*, all used directly in Chapter 5. The regime-dating success here is the template for asking, in Chapter 5, whether each MS-GARCH model's high-volatility regime lines up with known crisis episodes.

3 The models and the path dependence problem (Chapter 3)

3.1 The problem in one box

Path dependence

Take the most natural Markov-switching GARCH(1,1): every parameter switches with the regime,

$$\varepsilon_t = \sigma_t(S_{1:t})\,\eta_t, \qquad \sigma_t^2(S_{1:t}) = \omega_{S_t} + \alpha_{S_t}\varepsilon_{t-1}^2 + \beta_{S_t}\,\sigma_{t-1}^2(S_{1:t-1}).$$

Because the lagged variance on the right depends on the regime at $t-1$, which in turn depends on the variance at $t-2$ and the regime then, and so on, the variance at t depends on the *entire* path $S_{1:t} = (S_1, \ldots, S_t)$. With K regimes there are K^t possible paths. The likelihood contribution

$$f(r_t \mid \mathcal{F}_{t-1}) = \sum_{S_{1:t}} f(r_t \mid S_{1:t}, \mathcal{F}_{t-1})\,P(S_{1:t} \mid \mathcal{F}_{t-1})$$

therefore has K^t terms. At $t = 250$ with $K = 2$ that is 2^{250} terms, which is hopeless. Hamilton's filter needs the density to depend on only finitely many lagged regimes, and GARCH breaks that. The four responses:

- **Avoid:** drop the GARCH term and keep a finite-lag ARCH (Cai; Hamilton & Susmel).
- **Collapse:** replace $\sigma_{t-1}^2(S_{1:t-1})$ by an average over regimes (Gray; Klaassen).
- **Confront:** keep the model and use simulation-based estimation (Francq et al.; Francq & Zakoïan; Bauwens et al.; Augustyniak).
- **Remove:** redefine the model so that each regime has its own variance recursion (Haas et al.; Haas & Liu).

Branch 1: avoid it (MS-ARCH)

3.2 Cai (1994): Markov-switching ARCH

Cai, J. (1994). A Markov model of switching-regime ARCH. *Journal of Business & Economic Statistics*, 12(3), 309–316.

Used in: §3.2.

Question. Is the apparent unit-root persistence in interest-rate volatility caused by occasional shifts in the level of the variance? Cai formalises Diebold's suggestion.

Model. The mean switches with the regime and follows an AR(1). The variance is ARCH(q) with a regime-switching *intercept* only:

$$\sigma_t^2 = \omega_{S_t} + \sum_{m=1}^{q} a_m \varepsilon_{t-m}^2, \qquad S_t \in \{1, 2\}.$$

Cai explains why he does not use GARCH: with a GARCH term the estimation equations depend on the whole state history (2^T probabilities), which is infeasible for any realistic sample. He gives a covariance-stationarity proposition: the Markov persistence must be below one *and* the ARCH roots must lie outside the unit circle.

Evidence. Monthly excess return of the three-month over the one-month Treasury bill, August 1964–November 1991. The selected model is AR(1) with ARCH(2).

- **Single-regime:** ARCH(2) coefficients are 0.41 and 0.40, and GARCH(1,1) has $\alpha + \beta = 0.98$.
- **Switching model:** ARCH coefficients 0.06 and 0.09 (sum 0.14, insignificant). Staying probabilities are 0.99 (low) and 0.94 (high), so the Markov persistence is 0.93. The high-variance regime's asymptotic variance is more than ten times the low one's.

- Regime shifts are dated to the 1974 oil shock and the 1979–82 Federal Reserve policy change.
- No ARCH within regimes cannot be rejected.

Why it matters for the thesis

The cleanest early demonstration of the thesis question. Once regimes are allowed, persistence *moves* from the ARCH/GARCH terms (0.98) to the Markov chain (0.93), and the within-regime dynamics almost vanish. Costs to note for §3.2: GARCH is given up, and only the intercept switches. Note that Cai's data are monthly interest rates, not equities.

3.3 Hamilton & Susmel (1994): SWARCH

Hamilton, J. D., & Susmel, R. (1994). Autoregressive conditional heteroskedasticity and changes in regime. *Journal of Econometrics*, 64, 307–333. (Project file: Hamilton_and_Susmel_1993.pdf.)

Used in: §2.3 (the forecasting failure of GARCH), §3.2, and Chapter 5 (hand-coded SWARCH).

Question. ARCH models impute a lot of persistence to stock volatility yet forecast poorly. Could this be because large shocks such as October 1987 come from a different regime than small ones?

Model: SWARCH-L(K, q). Regimes *scale* an underlying ARCH process:

$$\varepsilon_t = \sqrt{g_{S_t}}\,\tilde{\varepsilon}_t, \qquad \tilde{\varepsilon}_t = \tilde{\sigma}_t \eta_t, \qquad \tilde{\sigma}_t^2 = \omega + \sum_{m=1}^{q} a_m \tilde{\varepsilon}_{t-m}^2 + \delta\, \mathbf{1}\{\tilde{\varepsilon}_{t-1} \le 0\}\, \tilde{\varepsilon}_{t-1}^2,$$

with $g_1 = 1$. Given the regimes, the variance of ε_t is $g_{S_t}\big[\omega + \sum_m a_m \varepsilon_{t-m}^2 / g_{S_{t-m}} + \dots\big]$, so it depends on $S_t, \dots, S_{t-q}$ only. The filter therefore runs over K^{q+1} configurations. The authors state explicitly that GARCH with a lagged-variance term cannot be accommodated in this framework.

- **Paper's notation:** P is column-stochastic, the transpose of ours; the leverage coefficient is ξ; and h_t is a standard deviation.

Evidence. Weekly returns on the NYSE value-weighted index, July 1962–December 1987 (T = 1,327), with an AR(1) mean.

- **GARCH benchmark:** Student-t GARCH-L(1,1) persistence is 0.96, so a shock still has weight $0.96^{52} = 0.12$ a year later. Its one-step forecast MSE (815.6) is *worse* than a constant variance (757.9).
- **Three regimes, SWARCH-L(3,2) with t:**
  - $g_2 = 4.4$ and $g_3 = 13.1$.
  - Staying probabilities are 0.9924, 0.9914 and 0.9831, giving expected durations of 132, 116 and 59 weeks.
  - The ARCH decay parameter is only 0.48.
  - The high-volatility regime covers the 1969–70 and 1973–75 recessions but not the brief 1980 one.
- **Four regimes:** a fourth regime ($g_4 = 169$) captures essentially only October 1987 and one week in January 1963. The Hessian was nearly singular. It is the only model to beat the constant variance in MSE (709.3).
- **Model selection:** Normal innovations are rejected in favour of t. AIC picks four regimes; Schwarz picks GARCH-L(1,1), then SWARCH-L(3,2).

Why it matters for the thesis

The central empirical precedent. Most persistence is attributed to long-lived regimes, and individual shocks die out within about a month. Two lessons for Chapter 5:

- Model selection criteria disagree (AIC versus BIC), so report both.
- Extra regimes can end up capturing single outliers.

The forecast evaluation here (MSE against a constant variance) is a good template.

Branch 2: collapse it

3.4 Gray (1996): generalized regime switching

Gray, S. F. (1996). Modeling the conditional distribution of interest rates as a regime-switching process. *Journal of Financial Economics*, 42, 27–62.

Used in: §3.3.

Question. Can a regime-switching model of the short rate keep full GARCH dynamics within regimes and still be estimable?

Model. The short rate has regime-specific mean reversion and a GARCH variance (the general form nests level effects), with transition probabilities that depend on the level of the rate. The path-dependence fix is to replace the lagged regime variance with the regime-*aggregated* variance of ε_{t-1}:

$$\sigma^2_{k,t} = \omega_k + \alpha_k \varepsilon^2_{t-1} + \beta_k \, \sigma^2_{t-1},$$
$$\sigma^2_{t-1} = \sum_j \xi_{j,t-1|t-2} \, \sigma^2_{j,t-1} \quad \text{(zero regime means)}.$$

Gray's version adds terms for the different regime means. Because σ^2_{t-1} does not depend on the path, the "tree" of variances recombines, and the likelihood has a first-order recursive structure like ordinary GARCH.

Evidence. Weekly one-month T-bill yields, January 1970–April 1994 (1,267 observations).

- **Single-regime GARCH:** $\alpha + \beta = 0.2095 + 0.8208 \approx 1.03 > 1$ *(sum computed; checked on page image)*, with poor residual diagnostics.
- **Regime-switching GARCH:** high-volatility regime $0.461 + 0.198 \approx 0.66$; low-volatility regime $0.166 + 0.269 \approx 0.43$ *(computed)*; staying probabilities 0.974 and 0.990.
- High-variance episodes: 1973–75, 1979–83 (the Fed experiment), late 1987.
- The GRS model outperforms single-regime models in several out-of-sample designs; constant-variance and GARCH forecasts sometimes produce negative R^2.

Why it matters for the thesis

Two points to take away:

- Persistence falls from above one to 0.43–0.66 once regimes are allowed, while regime persistence stays high.
- Gray explicitly argues that a single-regime model loads the persistence of regimes onto individual shocks.

Costs, argued by later papers: the regime parameters lose their clean interpretation (Haas et al.), no stationarity theory exists, and Gray's estimates are not consistent for the path-dependent model (Augustyniak). Interest rates, not equities.

3.5 Klaassen (2002): regime-switching GARCH forecasts

Klaassen, F. (2002). Improving GARCH volatility forecasts with regime-switching GARCH. *Empirical Economics*, 27, 363–394. (Project file: Klassen_2000_Improving. . . .pdf; the notes file is not a source.)

Used in: §3.3; Chapter 5 (hand-coded filter).

Question. GARCH volatility forecasts are too high in volatile periods. Is this because GARCH forces all persistence into individual shocks?

Model. Klaassen modifies Gray in two ways. The previous regime is integrated out at $t-1$ rather than $t-2$, which uses more data, and the integration conditions on the *current* regime:

$$\sigma^2_{k,t} = \omega_k + \alpha_k \varepsilon^2_{t-1} + \beta_k \sum_j P(S_{t-1} = j \mid \mathcal{F}_{t-1}, S_t = k)\, \sigma^2_{j,t-1},$$

$$P(S_{t-1} = j \mid \mathcal{F}_{t-1}, S_t = k) = \frac{p_{jk}\, \xi_{j,t-1|t-1}}{\sum_i p_{ik}\, \xi_{i,t-1|t-1}}.$$

Student-t innovations are used so that a single outlier does not trigger a regime switch. The likelihood is first-order recursive, and multi-step variance forecasts follow a convenient recursion, which is the paper's main technical contribution. In the paper, equation (4) is Gray's model and equation (5) is Klaassen's.

Evidence. Daily US-dollar rates for sterling, the Deutschmark and the yen, 3 January 1978–23 July 1997.

- **Constant-variance regimes:** the high regime's variance is 3–4 times the low regime's, and all staying probabilities exceed 0.975.
- **GARCH forecast efficiency:** the regression of realised on forecast variance rejects efficiency in 8 of 12 cases, with intercept > 0 and slope < 1 in all 12. High GARCH forecasts overstate the variance.
- **Regime-switching GARCH versus GARCH:** lower MSE in 11 of 12 cases (6 significant); relative gain in restricted R^2 of 22% at one day and 58% at ten days.
- Klaassen's model fits better than Gray's.
- **Nuance:** regime-switching *ARCH* beats GARCH in only 15 of 24 cases, and for sterling GARCH is best.

Why it matters for the thesis

The main evidence that regimes help *forecasting*, and the source for the forecast-efficiency regression and multi-step recursion you may use in Chapter 5. The 15-of-24 result suggests that regimes *without* GARCH are not enough, which is worth noting against Branch 1.

3.6 Kim (1994): switching state-space models and collapsing

Kim, C.-J. (1994). Dynamic linear models with Markov-switching. *Journal of Econometrics*, 60, 1–22.

Used in: §3.3 (antecedent of collapsing), Chapter 4/5 (smoother).

Contribution.

- Extends Hamilton's framework to general state-space models with Markov switching in both the measurement and transition equations.
- The exact filter would have to track a growing number of regime histories. Kim *collapses* the K^2 regime-conditional posteriors to K at each step, which makes filtering and likelihood evaluation approximate but fast.
- Presents an efficient algorithm for smoothed regime probabilities ($\xi_{k,t|T}$).
- Applied to Lam's generalised Hamilton model of GNP, the approximation performs well and saves considerable computing time.

Why it matters for the thesis

Two uses. Conceptually, it is the origin of "collapse the history at each step", the same trade Gray makes for GARCH (approximation for tractability). Practically, it is the smoothing algorithm behind the regime-dating plots in Chapter 5.

Branch 3: confront it (full path-dependent model; conceptual only)

3.7 Francq, Roussignol & Zakoïan (2001): GARCH driven by a hidden Markov chain

Francq, C., Roussignol, M., & Zakoïan, J.-M. (2001). Conditional heteroskedasticity driven by hidden Markov chains. *Journal of Time Series Analysis*, 22(2), 197–220.

Used in: §3.2 (MS-ARCH consistency), §3.4, Chapter 4.

Model. GARCH(p, q) with all coefficients depending on the hidden regime, which is the path-dependent model in the box above.

Main results.

- **Theorem 1:** a strictly stationary solution exists if and only if the top Lyapunov exponent γ of the associated random matrices is negative; it is then unique and ergodic.
- **Theorem 2:** second-order stationarity if and only if the spectral radius of a matrix built from the transition probabilities and regime coefficients is below one.
- For the **ARCH case** (no GARCH term) the MLE is shown to be consistent.
- A penalised likelihood is proposed for selecting the number of regimes and the ARCH order.

Evidence.

- **Simulation:** a single-regime GARCH fitted to data with two very different regimes gives estimates close to the nonstationary boundary ($\sum \alpha + \beta \approx 1$).
- **CAC 40 daily returns, June 1988–December 1993** (1,286 observations):
  - GARCH(1,1) is close to nonstationary.
  - Two-regime models have staying probabilities close to one and little heteroskedasticity within regimes.
  - Up to four regimes seem worth considering; five are rejected.

Why it matters for the thesis

The theoretical base of Branch 3, the origin of the Lyapunov-exponent approach used for IS-GARCH, and a second simulation (alongside Haas et al.) showing that ignoring regimes produces spurious near-unit persistence. Its MS-ARCH consistency result means Branch 1 has estimation theory that Branches 2 and 4 lack.

3.8 Francq & Zakoïan (2005): moments and ARMA structure

Francq, C., & Zakoïan, J.-M. (2005). The L^2-structures of standard and switching-regime GARCH models. *Stochastic Processes and their Applications*, 115, 1557–1582.

Used in: §3.4.

Main results. For the path-dependent MS-GARCH(p, q) with the chain independent of η_t:

- necessary and sufficient conditions for the existence of moments of any order, expressed through spectral radii of matrices built from P and the regime coefficients;

- ε_t^2 and its higher powers have the second-order structure of an ARMA process, with explicitly characterised orders. This holds also for standard GARCH, where it was previously known only for ε_t^2;
- a squared MS-GARCH is *not* in general a Markov-switching ARMA.

The paper itself proposes no estimator. It points to a companion paper, not in the folder, for moment-based estimation motivated by these results.

Why it matters for the thesis

Shows that the path-dependent model is theoretically well understood even though its likelihood is intractable: realism plus theory, but no easy estimation. Keep the discussion brief (it is first on the cut list for Chapter 3).

3.9 Bauwens, Preminger & Rombouts (2010): Bayesian MS-GARCH

Bauwens, L., Preminger, A., & Rombouts, J. V. K. (2010). Theory and inference for a Markov switching GARCH model. *The Econometrics Journal*, 13(2), 218–244.

Used in: §3.4. Cite this, not the 2006 discussion paper, which has a different model.

Model. Path-dependent MS-GARCH(1,1) in which both the mean and the GARCH parameters switch, driven by a hidden Markov chain with constant transition probabilities.

Main results.

- **Stationarity:** sufficient conditions for geometric ergodicity and existence of moments. The Nelson stability condition need only hold *on average* across regimes, weighted by the stationary probabilities. So some regimes can be integrated or even explosive while the process remains stationary.
- **Estimation:** ML is infeasible because of path dependence. Instead the regimes are treated as parameters and sampled by Gibbs sampling (griddy-Gibbs or Metropolis–Hastings for the GARCH parameters). Simulations show the sampler recovers the true parameters.

Evidence. Daily S&P 500 returns, 19 July 2001–20 April 2007 (1,500 observations).

- MS-ARCH does not capture enough persistence; MS-GARCH is preferred by a BIC-type criterion.
- In MS-GARCH the low-volatility regime's GARCH coefficients were poorly identified and were set to zero (constant variance).
- The high-volatility regime has $\alpha_2 = 0.054$ and $\beta_2 = 0.919$, so $\alpha_2 + \beta_2 = 0.973$ (posterior means). Staying probabilities are 0.978 and 0.985.
- Single-regime GARCH persistence is 0.990.

Why it matters for the thesis

The closest precedent to your data. Persistence splits into a Markov part and a within-regime part (0.973 in one regime, zero in the other), against 0.990 for single-regime GARCH. Two practical warnings for Chapter 5:

- Low-volatility GARCH parameters may be weakly identified.
- Restrictions such as $\alpha_1 = \beta_1 = 0$ may be needed.

3.10 Augustyniak (2014): maximum likelihood for the path-dependent model

Augustyniak, M. (2014). Maximum likelihood estimation of the Markov-switching GARCH model. *Computational Statistics & Data Analysis*, 76, 61–75.

Used in: §3.3 (critique of Gray), §3.4.

Question. Can the exact MLE of the path-dependent MS-GARCH be computed without modifying the model?

Method. A Monte Carlo EM algorithm, in which a Gibbs sampler draws regime paths in the E-step, combined with importance sampling (MCML) to reach the MLE and its asymptotic covariance. The log-likelihood for model comparison is approximated with a particle filter.

Main results.

- **Simulation:** used as an estimator of the path-dependent model, Gray's model gives biased estimates of ω_k and β_k, and the bias does not shrink with the sample size, so Gray's model does not estimate MS-GARCH consistently. It is, however, a good starting value.
- **Computation:** about 35 minutes (weekly) and 110 minutes (daily) per estimation.

Evidence. Weekly S&P 500 returns (October 1987–October 2012, 1,305 observations) and daily S&P 500 returns (20 May 1999–25 April 2011, 3,000 observations). The daily results (Table 8) use a constrained MS-GARCH with switching means and intercepts but common α and β:

- **GARCH:** $\alpha + \beta = 0.0759 + 0.916 \approx 0.99$ *(computed).*
- **Constant-variance MS:** staying probabilities 0.989 and 0.979.
- **MS-GARCH:** $\alpha = 0.0337$, $\beta = 0.942$ (≈ 0.976), staying probabilities 0.980 and **0.638**.
- **BIC:** MS-GARCH 4483.0 < Gray 4495.5 < GARCH 4510.2 < constant-variance MS 4661.8.

Adding GARCH dynamics makes the high-volatility regime much less persistent. For weekly data the unconstrained fit has $\beta_2 > 1$ yet is covariance stationary.

Why it matters for the thesis

Perhaps the most important single paper for the central question. On daily S&P 500 data, where persistence is attributed depends on the specification: with constant regime variances the chain is highly persistent, but once GARCH is added the high-volatility regime becomes short-lived and the persistence returns to $\alpha+\beta$. It also delivers the strongest critique of the collapse branch. Caveat: the comparison uses a constrained model.

Branch 4: remove it (IS-GARCH)

3.11 Haas, Mittnik & Paolella (2004): the focal model

Haas, M., Mittnik, S., & Paolella, M. S. (2004). A new approach to Markov-switching GARCH models. *Journal of Financial Econometrics*, 2(4), 493–530.

Used in: §2.4 (simulation), §3.5, Chapter 4, Chapter 5.

Critique of collapsing. In a GARCH model the ARCH(∞) form gives α the meaning of shock impact and β the meaning of memory. In Gray's model the regime variance responds to the *aggregated* variance, which moves with the filtered probabilities. Regime variances therefore change even when every $\alpha_k = 0$, and β_k can no longer be read as the memory of regime k.

Model (IS-GARCH). Every regime runs its own GARCH recursion, driven by the observed shock, in parallel:

$$\varepsilon_t = \sigma_{S_t,t}\,\eta_t, \qquad \sigma^2_{k,t} = \omega_k + \alpha_k \varepsilon^2_{t-1} + \beta_k \sigma^2_{k,t-1}, \quad k = 1, \ldots, K.$$

No regime history is needed, so the ordinary K-state filter gives the exact likelihood. In vector form the regime variances are $\boldsymbol{\sigma}^2_t$. In the paper, the regime is Δ_t, α_{1k} is our α_k, and δ is Markov persistence.

Main results.

- **Covariance stationarity** holds if and only if $\rho(M) < 1$, where M is a $K^2 \times K^2$ matrix. For $K = 2$ it is

$$M = \begin{pmatrix} p_{11}(\alpha_1 + \beta_1) & 0 & p_{21}(\alpha_1 + \beta_1) & 0 \\ p_{11}\alpha_2 & p_{11}\beta_2 & p_{21}\alpha_2 & p_{21}\beta_2 \\ p_{12}\beta_1 & p_{12}\alpha_1 & p_{22}\beta_1 & p_{22}\alpha_1 \\ 0 & p_{12}(\alpha_2 + \beta_2) & 0 & p_{22}(\alpha_2 + \beta_2) \end{pmatrix}.$$

  $\beta_k < 1$ for all k is necessary, but $\alpha_k + \beta_k < 1$ is *not*: a regime can be locally nonstationary.
- Closed forms for the unconditional variance and a fourth-moment condition ($\rho(Q) < 1$); the autocorrelation function of ε_t^2.
- If only ω switches, autocorrelations decay at rate $\max\{\alpha + \beta,\ p_{11} + p_{22} - 1\}$.
- **Simulation:** data are generated with switching ω, $\alpha + \beta = 0.75$, p from 0.5 to 0.995, $T =$ 10,000 and 2,500 replications. A single-regime GARCH's $\hat{\alpha} + \hat{\beta}$ increasingly absorbs the Markov persistence once it exceeds $\alpha + \beta$.
- **Special cases:** MS-ARCH (Francq et al.) and the mixed-normal GARCH with independent regimes.

Evidence. Daily US-dollar rates for the yen and sterling (January 1978–June 2003) and the Singapore dollar (January 1981–June 2003).

- All mixture models find at least one regime with $\alpha_k + \beta_k > 1$, with more weight on α_k.
- For the yen and sterling there is **no shift of persistence from GARCH to the chain**; Markov persistence is even insignificant in the two-regime models.
- The best overall model is the *independent*-mixture GARCH with skewed mixed-normal components; skewness mattered for density and VaR forecasts. The Markov version fits the Singapore dollar somewhat better.
- The authors conclude that normal densities *within* regimes may suffice.

Why it matters for the thesis

The focal model, and the paper to know best. Its strengths are the ones the thesis cites: exact likelihood, interpretable parameters, and a stationarity condition. Read the empirical section carefully for balance, because its own evidence does *not* show persistence migrating to the chain for two of three series. It favours normal (not t) regimes and an independent (not Markov) mixture. The locally nonstationary regimes it finds are excluded by MSGARCH's default constraints (see Ardia et al.).

3.12 Haas & Liu (2015): theory and a multivariate extension

Haas, M., & Liu, J.-C. (2015). *Theory for a multivariate Markov-switching GARCH model with an application to stock markets* [Conference paper]. Beiträge zur Jahrestagung des Vereins für Socialpolitik 2015, No. B22-V2. ZBW. https://hdl.handle.net/10419/112855

Used in: §3.5, Chapter 4, possibly Chapter 6.

Model. A multivariate version of IS-GARCH with regime-specific GARCH dynamics, leverage, and constant correlations within each regime (the regime-specific correlation matrices switch).

Main results.

- **Theorem 1:** a unique strictly stationary ergodic solution exists if and only if the top Lyapunov exponent is negative, generalising Liu (2006).
- **Theorem 2:** moment and covariance-stationarity conditions via spectral radii.

- **Theorem 3: MLE consistency under Gaussian innovations**. The authors note this was unknown even for the univariate Haas et al. model.
- An LM test for within-regime correlation dynamics, and a recursion for multi-step covariance matrices.

Evidence. Weekly MSCI World and FTSE EPRA/NAREIT global real-estate returns, January 1990–October 2011 ($T = 1{,}137$).

- BIC prefers two regimes and Student-t innovations, and t dominates Gaussian in and out of sample.
- The high-volatility regime has higher correlations.
- **Regime durations depend on the innovation distribution:** with Gaussian regimes, just over (under) one year for the low- (high-) volatility regime; with t regimes, almost five and four years. Gaussian regimes switch whenever an unusual observation occurs.

Why it matters for the thesis

The main theory reference for Chapter 4, with a key limit: consistency is proved only for *Gaussian* innovations, and asymptotic normality is not given, so the Student-t focal model lacks this backing. The duration result is central to your question: the innovation distribution changes how much persistence is attributed to the regimes. Also relevant if Chapter 6 needs a bivariate model.

4 Theory for IS-GARCH (Chapter 4)

Two kinds of stationarity

Chapter 4 relies on the distinction between *covariance* (second-order) stationarity and *strict* stationarity.

- **Covariance stationarity** requires a finite unconditional variance. For GARCH(1,1) the condition is $\alpha + \beta < 1$; for IS-GARCH it is $\rho(M) < 1$.
- **Strict stationarity** only requires the distribution to be time-invariant and can hold with infinite variance. It is characterised by a negative top Lyapunov exponent γ. For GARCH(1,1) the condition is $E[\ln(\beta + \alpha\eta_t^2)] < 0$.

IGARCH is the standard example of a process that is strictly but not covariance stationary. QMLE theory for GARCH typically needs only strict stationarity.

4.1 Nelson (1990): stationarity and persistence of GARCH(1,1)

Nelson, D. B. (1990). Stationarity and persistence in the GARCH(1,1) model. *Econometric Theory*, 6(3), 318–334.

Main results.

- With $\omega > 0$, GARCH(1,1) is strictly stationary and ergodic if and only if $E[\ln(\beta + \alpha\eta_t^2)] < 0$. By Jensen's inequality this holds whenever $\alpha + \beta \leq 1$, so **IGARCH(1,1) with $\omega > 0$ is strictly stationary**, even though its variance is infinite.
- With $\omega = 0$, $\sigma_t^2 \to 0$ almost surely in the IGARCH case.
- For IGARCH, the expected future variance grows linearly: $E(\sigma_{t+h}^2 \mid \mathcal{F}_t) = \sigma_{t+1}^2 + (h-1)\omega$ (Nelson's eq. 21, re-indexed).
- **"Persistence" depends on its definition.** In IGARCH, shocks persist in L^p for $p \geq 1$ (the Engle–Bollerslev sense) but *not* almost surely or in probability.
- Conditions for the existence of fractional moments.

Why it matters for the thesis

Two uses. In Chapter 4 it is the single-regime version of the strict-stationarity condition that Liu (2006) extends to IS-GARCH. In Chapter 2 it is a caution: $\alpha + \beta$ near or at one does not mean the process is explosive, and "persistence" should be defined precisely (the thesis uses forecast decay, the L^1 sense).

4.2 Bougerol & Picard (1992): the Lyapunov-exponent condition

Bougerol, P., & Picard, N. (1992). Stationarity of GARCH processes and of some nonnegative time series. *Journal of Econometrics*, 52, 115–127.

Main result. Write GARCH(p, q) as a random-coefficient vector autoregression $X_t = A_t X_{t-1} + B_t$ with i.i.d. nonnegative matrices. A strictly stationary solution exists if and only if the top Lyapunov exponent

$$\gamma = \lim_{t\to\infty} \tfrac{1}{t}\, E \ln\|A_t A_{t-1} \cdots A_1\|$$

is negative. For $p = q = 1$ this reduces to Nelson's condition. The same result is proved for generalised nonnegative autoregressions.

Why it matters for the thesis

The technique behind every strict-stationarity result in the folder: Francq et al. (2001), Liu (2006) and Haas & Liu (2015) all rewrite their model in this random-matrix form and apply the same logic. Chapter 4 should present it once, then specialise it.

4.3 Liu (2006): stationarity of IS-GARCH

Liu, J.-C. (2006). Stationarity of a Markov-switching GARCH model. *Journal of Financial Econometrics*, 4(4), 573–593.

Main results (for the Haas et al. model):

- **Theorem 1:** a unique weakly stationary solution exists if and only if $\rho(M) < 1$. This proves Haas et al.'s condition *without* their assumption that the process started in the infinite past with finite variance, which Liu calls uncheckable. It also gives an explicit causal expansion.
- Conditions for the existence of any integer moment of ε_t^2.
- **Theorem 3:** a unique strictly stationary ergodic solution exists if and only if the top Lyapunov exponent is negative (variance possibly infinite).
- **Corollary:** strictly stationary *with finite variance* if and only if $\rho(M) < 1$.
- **Integrated case** $\rho(M) = 1$: under further conditions, a unique strictly stationary solution with infinite variance, the analogue of IGARCH for IS-GARCH.

Why it matters for the thesis

The core stationarity reference for Chapter 4, and the reason the thesis can state both conditions ($\rho(M) < 1$ and $\gamma < 0$) for the focal model. The integrated case is a natural bridge back to the persistence puzzle: a regime-switching analogue of near-unit persistence.

4.4 Francq & Zakoïan (2004): QMLE for GARCH

Francq, C., & Zakoïan, J.-M. (2004). Maximum likelihood estimation of pure GARCH and ARMA-GARCH processes. *Bernoulli*, 10(4), 605–637.

Main results for the Gaussian QMLE of GARCH(p, q), single-regime:

- **Strong consistency** requires a compact parameter space, strict stationarity ($\gamma < 0$), η_t^2 non-degenerate with $E\eta_t^2 = 1$, and identifiability. It needs *no* moment condition on ε_t and allows the true value on the boundary.
- **Asymptotic normality** additionally requires the true parameter to be interior and $E\eta_t^4 < \infty$. The asymptotic variance is $(E\eta_t^4 - 1)$ times the inverse of a matrix built from derivatives of $\ln \sigma_t^2$. A finite variance of ε_t is still not needed.
- Boundary cases (e.g. true $\alpha = 0$) break asymptotic normality.
- The results extend to ARMA-GARCH, with extra moment conditions.

Why it matters for the thesis

The benchmark for what "QMLE theory" means in Chapter 4, and a checklist for what is established for IS-GARCH:

- Consistency: Gaussian case only, Haas & Liu (2015).
- Asymptotic normality: not in the folder.
- Student-t QMLE: not covered.

Note also the boundary issue. Low-volatility regimes with $\alpha_k \approx 0$ (as in Bauwens et al.) sit on

the boundary, where standard errors are unreliable.

4.5 Newey & Steigerwald (1997): consistency of non-Gaussian QMLE

Newey, W. K., & Steigerwald, D. G. (1997). Asymptotic bias for quasi-maximum-likelihood estimators in conditional heteroskedasticity models. *Econometrica*, 65(3), 587–599.

Main results.

- Consistency of any QMLE requires that the expected quasi-log-likelihood be maximised at the true conditional mean and *relative scale* parameters.
- For a non-Gaussian quasi-likelihood such as Student-t, this holds if the conditional mean is identically zero, *or* if both the assumed and the true innovation densities are unimodal and symmetric about zero.
- Without symmetry an extra location parameter is needed. Adding it when it is not needed has an efficiency cost, which they calculate; for the GARCH variance parameters there is *no* efficiency loss.

Why it matters for the thesis

Directly relevant to the focal Student-t specification. If the true S&P 500 innovations are asymmetric (which is likely), a t-based QMLE with a nonzero mean may be inconsistent, and the t likelihood identifies variance parameters only up to scale. Chapter 4 should state this, and Chapter 5 should report the t results alongside the Gaussian ones for this reason as well.

4.6 Davies (1977): nuisance parameters under the alternative

Davies, R. B. (1977). Hypothesis testing when a nuisance parameter is present only under the alternative. *Biometrika*, 64(2), 247–254.

Main results.

- **The problem:** when a parameter enters the model only under the alternative, standard LR, Wald and score tests do not have their usual χ^2 distributions.
- **The reduction:** large-sample theory reduces the problem to inference about the maximum of a Gaussian process indexed by the nuisance parameter. The proposed test rejects when the supremum of the statistic exceeds a critical value, and approximate bounds for the significance level and power are derived.

Why it matters for the thesis

The "Davies problem" for Chapter 4. When testing one regime against two, the transition probabilities and second-regime parameters are not identified under the null, so the likelihood-ratio statistic comparing GARCH with IS-GARCH is not χ^2. This is why the thesis cannot simply use LR tests for the number of regimes, and why information criteria are used in practice.

4.7 Cho & White (2007): testing for regime switching

Cho, J. S., & White, H. (2007). Testing for regime switching. *Econometrica*, 75(6), 1671–1720.

Main results.

- **Approach:** a quasi-likelihood ratio (QLR) test of one versus two regimes that uses a *mixture* likelihood and deliberately ignores the Markov serial dependence of the regimes. The resulting test is sensitive to the mixture aspect and potentially powerful against switching that is not Markov.

- **Nonstandard asymptotics:** the setting has nuisance parameters under the alternative, parameters on the boundary, and higher-order approximations. They derive null distributions under β-mixing data, and show that ignoring the boundary gives overly conservative critical values.
- **Practical tools:** critical values for popular models and bounds for conservative ones. Monte Carlo evidence shows good size and power.
- **Application:** Porter's (1983) cartel data, where the original conclusions are confirmed.

Why it matters for the thesis

The candidate formal test for Chapter 4. **Caution:** the paper does not treat GARCH-type conditional variances. Whether its results carry over to IS-GARCH is not established in the folder, so any application to the thesis models should be flagged as an extension or an informal check.

5 Software (Chapter 5)

5.1 Ardia et al. (2019): the MSGARCH package

Ardia, D., Bluteau, K., Boudt, K., Catania, L., & Trottier, D.-A. (2019). Markov-switching GARCH models in R: The MSGARCH package. *Journal of Statistical Software*, 91(4), 1–38. https://doi.org/10.18637/jss.v091.i04

What it provides.

- **Models:** implements the Haas et al. (2004) Markov-switching specification and the Haas et al. mixture-of-GARCH specification. Regime variances can be sARCH, sGARCH, eGARCH, gjrGARCH or tGARCH.
- **Distributions:** normal, Student-t (labelled "std", with $\nu > 2$) and GED, plus Fernández–Steel skewed versions.
- **Estimation and output:** ML and Bayesian MCMC estimation; simulation; one- and multi-step density forecasts; VaR and expected shortfall; smoothed and filtered probabilities.
- **Constraints:** parameters can be restricted to be equal across regimes (e.g. a common β).

Important defaults.

- The data are assumed to have zero mean, so returns must be demeaned first.
- Positivity and **covariance stationarity are imposed in each regime** (e.g. $\alpha_k + \beta_k < 1$). The authors note this is stronger than Haas et al.'s $\rho(M) < 1$.

Why it matters for the thesis

Your main estimation tool, with two consequences to decide on (see Section 7):

- The per-regime constraint rules out the locally nonstationary regimes Haas et al. found.
- The zero-mean assumption interacts with Newey & Steigerwald's consistency conditions for the t likelihood.

6 Synthesis: where persistence went

The table collects, paper by paper, how much persistence a single-regime model attributes to shocks and what happens when regimes are allowed. Entries marked (c) are my computations from reported estimates.

Paper	Data	Single-regime persistence	With regimes
Bollerslev (1986)	US inflation, quarterly	$\alpha + \beta \approx 0.96$ (c)	—
Bollerslev (1987)	FX daily; S&P 500 monthly	0.978, 0.976; 0.842 (c)	— (frequency matters)
Lamoureux & Lastrapes (1990a)	20 stocks, daily	mean 0.728	not regimes: 0.073 once volume is added
Cai (1994)	T-bill excess, monthly	0.98 (GARCH)	ARCH sum 0.14; Markov persistence 0.93
Hamilton & Susmel (1994)	NYSE, weekly	0.96 (t-GARCH-L)	ARCH decay 0.48; regimes last 59–132 weeks
Gray (1996)	T-bill yields, weekly	≈ 1.03 (c)	0.66 and 0.43 within regimes (c); staying probabilities 0.97, 0.99
Francq et al. (2001)	CAC 40, daily	near nonstationary	staying probabilities near 1; little within-regime ARCH
Klaassen (2002)	FX, daily	high	staying probabilities > 0.975; better forecasts
Haas et al. (2004)	FX, daily	high	some regime with $\alpha_k + \beta_k > 1$; **no** shift to the chain for yen and sterling
Bauwens et al. (2010)	S&P 500 daily 2001–07	0.990	0.973 in high regime, 0 in low; staying probabilities 0.978, 0.985
Augustyniak (2014)	S&P 500 daily 1999–2011	≈ 0.99 (c)	MS: staying probabilities 0.989, 0.979; MS-GARCH: 0.976 (c) and $p_{22} = 0.638$
Haas & Liu (2015)	world equity and real estate, weekly	—	regime durations about 1 year (Gaussian) versus 4–5 years (t)

Reading the table. The evidence is *not* uniform, and this is good news for the thesis question.

- **Constant variance within regimes** (Cai; Hamilton & Susmel; the MS columns of Augustyniak): the chain absorbs the persistence.
- **GARCH within regimes** (Augustyniak; Haas et al.): persistence can move back into the GARCH terms, and regimes become short-lived or insignificant.
- **Innovation distribution** (Haas & Liu; Klaassen): moving from normal to t lengthens regimes.

So "where persistence lives" depends on the specification, which is exactly the hypothesis the empirical chapter can test.

7 Open issues raised by the reading

1. **MSGARCH's per-regime stationarity constraint** (Ardia et al.) excludes the locally nonstationary regimes found by Haas et al. (2004) and allowed by Bauwens et al. (2010). Decide whether to accept it or hand-code IS-GARCH constrained only by $\rho(M) < 1$.
2. **Theory gap for the focal model:** consistency is proved only for Gaussian innovations (Haas & Liu, 2015); asymptotic normality and the Student-t QMLE are not covered; and Newey & Steigerwald (1997) show when a t QMLE can be inconsistent.

3. **Testing the number of regimes:** the Davies problem rules out standard LR tests, and Cho & White (2007) does not cover GARCH. Information criteria will do most of the work; say so explicitly.
4. **Normal versus t within regimes:** Haas et al. (2004) favour normal regimes, while Haas & Liu (2015) and Klaassen (2002) favour t. Since the distribution changes where persistence is attributed, estimating both is essential, and the thesis already plans this.
5. **Balance on the focal model:** Haas et al.'s own evidence finds no shift of persistence to the chain for two of three series, and prefers the independent-mixture version. This should be reported in §3.5.
6. **Hedging (Chapter 6):** if a bivariate model is needed, Haas & Liu's multivariate specification is the natural candidate. Decide before §3.5 is drafted.

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